

Day 6 Introduction to Network device theory

1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Device	Main Function	Typical Ports / Interfaces
Switch	Connects multiple wired devices in the same LAN and forwards Ethernet frames using MAC addresses.	Ethernet ports such as RJ45 (1–48 ports) ; management may use 22 (SSH) or 23 (Telnet)
Router	Connects the home LAN to the Internet, routes IP packets, and commonly provides DHCP and NAT.	WAN/LAN Ethernet ports ; services may use 80/443 for management, 67/68 UDP for DHCP
Hub	Connects multiple wired devices and broadcasts incoming data to all ports.	Ethernet/RJ45 ports; operates mainly at Layer 1 and does not use TCP/UDP ports
Firewall	Filters incoming and outgoing network traffic according to security rules and helps protect the LAN from unwanted connections.	Usually integrated into the router; commonly manages traffic on 80/443 , plus other TCP/UDP ports depending on services
Access Point (AP)	Provides Wi-Fi connectivity and connects wireless devices to the wired LAN.	Ethernet uplink; Wi-Fi uses 2.4/5/6 GHz radio bands rather than TCP/UDP ports; management may use 80/443

2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

Device	Port / Component	Function
1. Router	Ethernet LAN ports	Connect computers and other wired devices
	WAN/Internet port	Connects the router to the modem/Internet
	Console port	Used for configuration and management on some routers
	Power port	Supplies electrical power
	Antenna	Provides wireless/Wi-Fi connectivity
2. Switch	Ethernet ports	Connect PCs, printers, routers, and other wired devices

Device	Port / Component	Function
	Console port	Used for switch configuration on managed switches
	Power port	Supplies electrical power
3. Wireless Access Point	Ethernet port	Connects the AP to the wired network
	Power port / PoE	Supplies power to the AP
	Antenna	Sends and receives Wi-Fi signals
	Console port	Available on some enterprise APs for configuration

3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.
use that app.

Example: Instagram

I use Instagram daily through my phone connected to my home Wi-Fi.

Access Point: Connects my phone to the home Wi-Fi network and sends my Instagram data to the local network.

Router: Routes the data from my home network to the Internet and forwards Instagram's responses back to my phone.

Firewall: Helps filter unwanted or unauthorized network traffic between my home network and the Internet.

4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

College Wi-Fi Problem

One possible reason is that the **Wi-Fi access point (AP) is overloaded** because too many students are connected to the same AP at the same time. This can cause slow speeds and frequent disconnections.

Solution: Upgrade to a higher-capacity access point or install additional APs to distribute users across multiple APs. This improves **coverage, device capacity, and traffic handling**, resulting in a more stable Wi-Fi connection.